

National Entrepreneurship Challenge (NEC)

E-Cell, IIT Bombay

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My Motivation

My drive toward entrepreneurship stems directly from bench-testing hardware. Last semester, when building a simple water quality sensor for rural borewells, I spent three weeks troubleshooting a noisy ADC pin only to realize the issue was component quality from local vendors. That frustration taught me something vital: engineering a working prototype on a breadboard is barely 10% of the battle. The remaining 90% is component sourcing, board spin costs, unit economics, and building something robust enough for real field conditions.

I don't view entrepreneurship as a fancy title--to me, it's the grit to debug a physical hardware product until it actually solves a problem for someone who needs it. I enjoy diving into circuit design, microcontroller firmware, and mechanical enclosures, but I've come to realize that technical skills alone can't sustain a company if you don't understand cash flows and vendor negotiation.

My Expectations from E-Cell

I expect E-Cell to pull me out of my technical bubble and force me to confront the commercial realities of hardware startups. Specifically, I want practical guidance on hardware seed funding, prototyping grants, and navigating electronic supply chains in India. Most textbooks skip over how small hardware teams manage minimum order quantities (MOQs) with fab houses or how to protect IP before hitting the market.

Through NEC, I hope to get blunt feedback from founders who have actually shipped physical products in India. I want to learn how to pitch complex technical architectures to investors without losing them in schema details, and build a network of mentors who can hold our team accountable to tight delivery milestones.